

MARKET PROBLEM & OPPORTUNITY BRIEF

The Auction Tax

The hidden daily cost of open & close order exposure in equities — and the on-chain opportunity for private, verifiable price formation.

Every trading day, the two most important moments in the market — the open and the close — are also its most exposed. Institutional-size orders must reveal their intent before they execute, and systematic participants price that information in seconds. As trillions in real-world assets tokenise, the same failure is being rebuilt on transparent blockchains, where the leakage is total.

~10%

of US-equity daily volume prints in the closing auction

\$18.9B

average daily value in NYSE's closing auction alone

~5.5 bps

average price move the instant imbalance data is disclosed

\$5–50M+

estimated daily US-equity auction execution cost (working estimate)

\$0

private, verifiable open/close auctions that exist on-chain today

CrossDesk — a sealed, uniform-price auction that produces the official open/close (NAV) price for tokenised assets on Canton.

Prepared July 2026 · Sebastian Borjas · github.com/superbigroach/canton-dvp-settlement-desk

Figures are sourced working estimates for framing, not precise facts. See Sources & Method.

Executive summary

Market open and close auctions concentrate a huge share of daily liquidity into a few seconds — and require participants to **reveal order intent before the price prints**. That disclosure is efficiently exploited: the moment an imbalance is published, prices move ~5.5 bps and professional traders reprice within 300 milliseconds. The result is a recurring, structural **"auction tax"** on the very orders that most need a fair price: institutional size and, increasingly, the tokenised funds that mark their NAV to the close.

~10% of US-equity daily volume · ~\$18.9B/day on NYSE alone

transacts in the closing auction — the single largest liquidity event of the day, and the one most exposed to information leakage.

Sources: BMLL / Databento; NYSE.

The tokenised parallel is worse. As real-world assets move on-chain (tokenised money-market funds, USDCx, wrapped ETH/BTC), they inherit the need for an official open/close price — but on transparent blockchains the order book and mempool are *fully* public, turning leakage into industrialised front-running (MEV). **No private, verifiable open/close auction exists on-chain today.**

CrossDesk closes that gap: a sealed, uniform-price call auction on the Canton Network that runs the opening and closing cross, keeps every order private until the print, clears at a single deterministic price computed on-ledger, and settles each fill atomically. The clearing price *is* the official open/close — a benchmark that cannot be gamed before it exists.

1 · The problem: the open and close are where price integrity breaks

Auctions exist because continuous trading is bad for size. Concentrating liquidity into a single cross gives large orders a fair, single price and produces the market's **official benchmark** — the print that funds, indices, and derivatives settle against. This is why the close has become the gravitational centre of the trading day: **roughly 10% of US-equity average daily volume now transacts on the close**, and the share keeps growing as passive and index strategies route to the benchmark.

Sources: Databento; BMLL; State Street Global Advisors, "Closing time: how passive investing is reshaping equity market microstructure."

The structural flaw: you must show your hand before you trade

To participate in the cross, orders are entered into an auction book *while continuous trading is still live*, and exchanges publish the resulting **order imbalance** to all professional participants before the print. That disclosure is the vulnerability:

- **Imbalance is a signal.** A large buy imbalance tells the market a price-insensitive buyer *must* execute — and the print will likely be higher.

- **The market reprices instantly.** Closing prices shift **~5.5 bps on average the moment imbalance data is released**, and professionals reprice within ~300ms.
- **The original order pays for it.** The mover has revealed intent, guaranteed execution, and now fills at a worse, anticipated price — classic adverse selection.

Source: Nasdaq, "How much does the MOC imbalance matter?"

First-hand perspective. The founder previously traded institutional equity volume across North American and European markets. That experience is direct: volatility and price dislocation concentrate visibly into the open and close, and systematic desks are built specifically to read auction imbalance and position ahead of the print. The "auction tax" described here is not theoretical — it is a daily, observable feature of how these sessions trade.

2 · Quantifying the cost

There is no single published figure for "money lost to open/close leakage," because most of the cost is legitimate market impact and adverse selection rather than provable front-running. But the *addressable* cost — the execution penalty created specifically by exposing order intent before the print — can be bounded.

Component	Figure	Source / basis
US closing-auction share of ADV	~10%	Databento; BMLL
NYSE closing auction, avg daily value	~\$18.9B	BMLL / NYSE
Price move on imbalance disclosure	~5.5 bps	Nasdaq
Institutional price impact, 1% ADV close trade	~8.2 bps	JFQA (Cambridge)
Est. US auction execution cost / day (working est.)	\$5–50M+	See method below
Annualised (≈250 trading days)	~\$1.25–12B+	Derived

Method (transparent, conservative)

US closing auctions transact tens of billions of dollars daily (NYSE alone ~\$18.9B; the consolidated close is materially larger). Applying even **1–5 bps of avoidable, leakage-driven execution cost** to that flow yields an order-of-magnitude of **~\$5M–\$50M+ per trading day** in US equities. This is deliberately conservative and excludes opening auctions, European and Asian sessions, and futures/ETF arbitrage that trades around the same prints.

Honest nuance: most of this is normal impact and price discovery — *not* illegal front-running, which is small and not publicly observable. The point is not to allege wrongdoing; it is that **exposing order intent before the print is a structural cost**, and a mechanism that removes the exposure removes the tax.

3 · Why it happens — and why it is getting bigger

The mechanics

- **Concentration.** Auction orders are large, time-clustered, and partially predictable.
- **Visibility.** Imbalance feeds broadcast net buy/sell pressure before the cross.
- **Anticipation.** HFTs, stat-arb, index-arb and ETF-arb desks position into the expected move.
- **Certainty of execution.** On-close orders *must* fill, so counterparties can trade against them with confidence.

The trend

- **Passive is eating the close.** Index and ETF flows route to the benchmark, growing auction share and concentration.
- **Fragility at scale.** On the June 2024 Russell reconstitution, NYSE and Nasdaq traded \$275B and \$103B in the final moments — enormous, exposed prints.
- **Benchmarks depend on it.** More NAVs, closes, and settlements reference a single, leak-prone price.

Sources: State Street Global Advisors; BMLL, "Into the close: unpacking US closing-auction dynamics and the Russell reconstitution."

4 · The same failure is being rebuilt on-chain — worse

Real-world assets are tokenising fast: bank-issued deposit tokens, tokenised money-market funds, and wrapped ETH/BTC are now live on institutional ledgers. Every one of them needs the same things equities need — **a fair way to execute size and an official open/close price to mark NAV against.**

On **transparent** blockchains, the leakage problem is not smaller — it is total. The entire order and pending-transaction set is public, so anticipatory trading becomes fully industrialised **MEV** (front-running and sandwiching). A large tokenised order on a continuous on-chain venue is not merely exposed by an imbalance feed; it is exposed byte-for-byte in the mempool.

The gap: there is **no private, verifiable open/close auction on-chain today.** Continuous DEXs leak; OTC gives no public price discovery; and tokenised funds have no trustworthy, un-gameable print to mark to. This is the whitespace CrossDesk occupies.

5 · The solution: CrossDesk

CrossDesk is a sealed, uniform-price call auction on the Canton Network that runs the opening and closing cross for tokenised assets. It removes the exposure that creates the auction tax:

The auction is the price-formation core, but CrossDesk is the fuller **fund-issuance + NAV layer**. The sealed cross produces a **credibly-neutral NAV struck by a K-of-N committee** — no single admin sets the price — and the same atomic DvP engine powers **in-kind ETF/basket create & redeem**: authorised participants deliver the underlyings and mint (or burn) shares atomically against that committee NAV.

The problem

How CrossDesk removes it

Order intent leaks before the print	Sealed book — each order is visible only to the venue and the trader who placed it, by construction (Canton sub-transaction privacy). Nothing to anticipate.
The reference price can be gamed	Deterministic clearing — one uniform price that maximises matched volume, computed on-ledger. The price cannot exist until the cross runs, so it cannot be traded ahead of.
Large orders move the market	Single-price cross with pro-rata rationing of the heavy side — size executes at one fair price, no impact.
Settlement risk / no audit trail	Atomic DvP — every fill settles all-or-nothing in one transaction with an immutable, need-to-know audit receipt.
No trustworthy on-chain NAV	The clearing price is the official open/close — a verifiable benchmark funds can mark to, struck by a K-of-N committee rather than any single admin.
No native, trustworthy on-chain fund issuance	In-kind create/redeem — deliver underlyings ↔ mint/burn shares atomically against the committee NAV, on the same DvP engine.

Why only Canton: a sealed order book is impossible on a transparent chain — the entire premise (hidden interest until the print) leaks. Canton's privacy makes the private auction possible; its atomic settlement makes the fill final; its permissioned identities make the audit trail compliant.

Who pays, and the market

- **Tokenised-fund / RWA operators** — pay for a private, auditable daily NAV auction (NAV-as-a-service).
- **Institutional desks & large holders** — pay a per-auction execution fee for impact-free, MEV-proof block execution.
- **Venues & custodians** — pay a platform fee to run a branded, compliant cross for members.
- **Tailwind** — tokenised RWAs are scaling into the tens of billions and require exactly this market structure, which does not yet exist.

Build status: CrossDesk is already built — Daml contracts with 18/18 tests green, a Spring backend over the Ledger API, and a React desk — and is **deployed and connected on the shared HackCanton devnet node (Canton 3.x)**: the backend connects over TLS + JWT and reads work end-to-end (writes are pending a node-operator permission grant).

Sources & method

Figures are sourced working estimates assembled for framing a problem and opportunity; they are not precise facts and should not be relied upon as investment or execution guidance. Cost estimates are order-of-magnitude and deliberately conservative.

1. Databento — "Nearly 10% of average daily volume transacts on the close ... the closing auction is one of the most important events of the trading day." x.com/DatabentoHQ
2. BMLL Technologies — "Into the Close: Unpacking U.S. Closing Auction Dynamics and the Impact of the Russell Reconstitution." bmlitech.com/news
3. NYSE — U.S. Equity Volumes & Data Insights, "Closing Auction: immediate market impact, price drift and transaction cost of trading." nyse.com/data-insights
4. Nasdaq — "How Much Does the MOC Imbalance Matter?" (closing prices shift ~5.5 bps immediately after imbalance data; ~300ms repricing). nasdaq.com/articles
5. Journal of Financial and Quantitative Analysis (Cambridge) — "Price Impact in Closing Auctions, Opening Auctions, and Continuous Markets: A Benchmark for Cost of Trading on Anomalies." jfq.org · cambridge.org/core

6. State Street Global Advisors — "Closing time: How passive investing is reshaping equity market microstructure." ssga.com
7. Market microstructure of order-imbalance behaviour — [arXiv:1707.05550](https://arxiv.org/abs/1707.05550); [arXiv:2012.10145](https://arxiv.org/abs/2012.10145) (heavy-tailed distributions in closing auctions).

Prepared by Sebastian Borjas, July 2026, for the CrossDesk project (HackCanton Season 2). This document is an independent market brief for evaluation and discussion. It is not investment advice, an offer, or a solicitation, and it makes no allegation of unlawful conduct by any market participant — the costs described arise from legitimate market structure. Canton and Daml are products of Digital Asset; cETH is a product of onRails; CBTC of BitSafe. Independent and unaffiliated. Repo: github.com/superbigroach/canton-dvp-settlement-desk